Matrix Diagonalization

Lecture 20260513 : For Matrix $A = \begin{bmatrix} 0'7 & 0'2 \\ 0'3 & 0'8 \end{bmatrix}$. Eigenvecs: $\vec{v_1}\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\vec{v_2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
Eigenvals: $\lambda_1 = 1$ $\lambda_2 = 0'5$

A matrix A is diagonalizable if we can write it $A = PDP^{-1}$

D: is diagonal with entries the eigenvals.

P: the columns of P are the eigenvecs of A.

ex:

$$A = \begin{bmatrix} 0'7 & 0'2 \\ 0'3 & 0'8 \end{bmatrix} \qquad P = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 0'5 \end{bmatrix}$$

$$P^{-1} = \frac{1}{\det P} \operatorname{Adj} P = \frac{1}{\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix}} \cdot \begin{bmatrix} -1 & -1 \\ -3 & 2 \end{bmatrix} = \frac{1}{-5}\begin{bmatrix} -1 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ \frac{3}{5} & \frac{-2}{5} \end{bmatrix}$$

$$P \cdot D \cdot P^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0'5 \end{bmatrix}\begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{bmatrix} = \ldots = A$$

Matrix Powers

If $A = PDP^{-1}$ then $A^2 = PD\cancel{P^{-1}P}DP^{-1} = PD^2P^{-1}$

In general $A^K = PD^KP^{-1}$

This is why diagonalization matters. It makes repeated application of the matrix manageable.

Logistics example: long-run hub distribution

Suppose the initial stock vector is $x_0 = \begin{bmatrix} 100 \\ 0 \end{bmatrix}$

We write it as a combination of the two eigenvectors:

$$x_0 = \alpha \vec{v}_1 + \beta \vec{v}_2 = \begin{bmatrix} 100 \\ 0 \end{bmatrix} = \alpha \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} \longrightarrow$$

$$\longrightarrow \begin{matrix} 100 = 2\alpha + \beta \\ 0 = 3\alpha - \beta \end{matrix} \rightarrow 3\alpha = \beta \rightarrow 100 = 2\alpha + 3\alpha \rightarrow \begin{matrix} \alpha = 20 \\ \beta = 60 \end{matrix}$$

$$x_0 = 20 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 60 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad t = 0$$

After k cycles of delivery $t = k$

$$x_k = 20 \cdot 1^k \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 60 \cdot 0'5^k \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 40 + 60 \cdot 0'5^k \\ 60 - 60 \cdot 0'5^k \end{bmatrix}$$

As $k \to \infty$, then $0'5^k \to 0$, so $x_{k\to\infty} \sim \begin{bmatrix} 40 \\ 60 \end{bmatrix}$

This is the long run distribution. The initial imbalance vanishes. The system converges to the eigenvector associated with the dominant eigval. 1.

$$x_{k\to\infty} \sim \begin{bmatrix} 40 \\ 60 \end{bmatrix} \quad // \quad \vec{v}_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \lambda_1 = 1$$

Symmetric matrices vs. non-symmetric matrices

If $A = A^T$, then A is symmetric.

For real ($\mathbb{R}$) symmetric matrices:

- ALL eigenvals are real
- eigevecs belonging to different eigenvals are orthagonal.
- The matrix is ALWAYS diagonalizable.

More precisely $A = Q \cdot D \cdot Q^T$ with Q orthogonal.

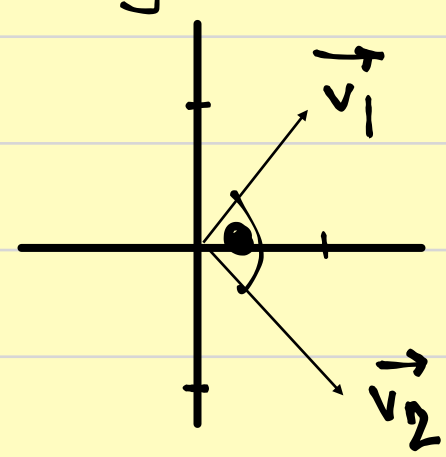
Ex: $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$\det[S - \lambda I] = \det\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

$\lambda_1 = 3 \quad \lambda_2 = 1$

$\downarrow \qquad \downarrow$

$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$



$$\vec{v}_1^T \cdot \vec{v}_2 = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot 1 + 1(-1) = 0$$

- If $A \neq A^T$ then the right and lef eigenvecs generally differ. The matrix may be diagonalizable, but not nelesarily an orthogonal matrix.

Strictly speaking: a straight line Euclidean distance matrix is symmetric. Real road travel time or road-network travel distance is always non-symmetric.

Ex:

$$T = \begin{array}{c} \\ \\ \\ \\ \end{array}\begin{array}{cc} \begin{bmatrix} 0 & 18 & 25 \\ 22 & 0 & 14 \\ 20 & 16 & 0 \end{bmatrix} & \begin{array}{c} A \\ B \\ C \end{array} \\ \begin{array}{ccc} A & \quad B & \quad C \end{array} & \end{array}$$

time $A \to B = 22$
time $B \to A = 18$

Not every matrix is diagonalizable.

A matrix may have eigenvalues, but still fail to have enough independent eigenvectors. Then the diagonalization is impossible.

Ex:

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\det\left[J - \lambda I\right] = \det\begin{bmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 \to \lambda_1 = \lambda_2 = 1$$

$(J - I)\vec{v} = \vec{0} \to \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \to$ The eigenvectors are multiples of $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

There is only one independant eigenvector.
So J is <u>not</u> diagonalizable.

Dominant eigenvalue interpretation

If a matrix is diagonalizable <u>and</u> one eigenval has strickly largest magnitude, then repeated powers of the matrix are dominated by the eigenvalue and its eigenvector.

$$A = \begin{bmatrix} 0{,}7 & 0{,}2 \\ 0{,}3 & 0{,}8 \end{bmatrix}$$

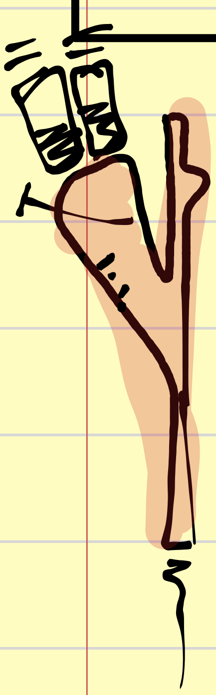
The dominant eigenvalue is $\lambda_1 = 1$.

" " eigenvector is $\vec{v_1} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

The long run stock ratio will be $2 : 3$.

Eigenvals & eigenvecs help logistics engineers analyze:

- repeated transport processes.
- long-run stock distributions.
- direction-dependent road & network relations
- stable vs. unstable modes
- dominant patterns in multi-location systems



KEY ENGINEERING IDEAS:

1. A right eigenvector gives a pattern preserved in direction.
2. A left eigenvector gives an important row-wise weighting (often a conservation law).
3. The dominant eigenvalue controls long-run

behaviour.

4. Diagonalization simplifies repeated application: $A^k = PD^kP^{-1}$

5. Symmetric & non-symmetric matrices behave differently.

6. Real world matrices are non-symmetric.

Ex. $x_k = \begin{bmatrix} 40 + 60 \cdot 0.5^k \\ 60 - 60 \cdot 0.5^k \end{bmatrix}$